Falu Hong

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Education

Nanjing University, Nanjing China

Master of Cartography and Geographic Information System **Sep 2017 - Jul 2020 (expected)**

- Advisor: Prof. Wenfeng Zhan (Professor for International Institute for Earth System Science, NJU)
- Overall GPA: 4.56 (91.2/100), Major GPA: 4.46 (89.2/100)
- Selected core courses: Modern physical geography, Basic issues of human geography, Geography information science, Remote sensing of atmosphere environment, Remote sensing physics, Thermal remote sensing: Theory and application, Remote sensing of landscape ecology

Nanjing University (NJU), Nanjing, China

Bachelor of Geography Information Science (GIS) **Sep 2013 - Jul 2017**

- Overall GPA: 4.43 (88.6/100), Major GPA: 4.43 (88.6/100)
- Selected core courses:

Math and programming	Geoscience	GIS	Remote sensing
Calculus	Human geography	Introduction to GIS	Introduction to remote sensing
Probability theory and mathematical statistics	Introduction to physical geography	Principles of geographic information systems	Digital image processing
Linear algebra	Principles of hydrology	Digital terrain modelling and analysis	Land surface remote sensing
The C programing language	Introduction to atmospheric sciences	GIS technology and application	Physical principles of remote sensing
Spatial data structure	Fundamental soil science	Design of geographical information systems	Geographical analysis for remote sensing

Research Interest

Thermal remote sensing, time-series analysis, land-atmosphere interaction, urban heat island, drought analysis, evapotranspiration, soil moisture, hydrology, climate variability and change, land use and land cover change

Research Experiences

Evaluations of diurnal land surface temperature cycle (DTC) models under clear-sky

Undergraduate thesis under the supervision of Prof. Wenfeng Zhan **Oct 2016 - May 2017**

- Obtain nine representative four-parameter DTC models with a set of parameter-reduction strategies.
- Comprehensive assessment of DTC models with geostationary satellite data and *in-situ* measurements.
- Provide the parameter-reduction order and the best-performance four parameter DTC model.

Design of a framework to generate physically true daily mean land surface temperature from thermal remote sensing satellite data

Supervised independent research **Jul 2017 - Sep 2018**

- Design a framework which combines annual temperature cycle and diurnal temperature cycle models to generate physically true daily mean land surface temperature.
- Evaluate the accuracy and sensitivity of the framework with geostationary satellite data under clear-sky condition and *in-situ* measurements under cloudy condition.
- Quantify the sampling bias caused by simply averaging sporadic clear-sky observations.

Fast generation of global daily mean 1-km land surface temperature products

Supervised independent research **May 2018- present**

- Generate global all-weather daily mean 1-km land surface temperature products by combining remote sensing data from polar-orbiters and reanalysis data.
- Employ downscaling technique to boost the calculation speed based on the new designed framework which combines annual temperature cycle and diurnal temperature cycle models.
- Analyze the global temporal trend of LST variations from 2002 to 2017.

Enhancement of the annual land surface temperature cycle (ATC) model

Cooperation | *results analysis, coding and paper revision* **Jun 2018 - Aug 2018**

- Enhance the traditional sinusoidal ATC model by consideration of short-term weather fluctuations with the assistance of LST-related controlling factors.
- Propose a hybrid and flexible framework to balance the prediction accuracy and generalization ability of ATC models.

Publications

[1] **Hong, F.**, Zhan, W., Göttsche, F.M., Liu, Z., Zhou, J., Huang, F., Lai, J., Li, M. (2018). Comprehensive assessment of four-parameter diurnal land surface temperature cycle models under clear-sky. *ISPRS Journal of Photogrammetry and Remote Sensing*, 142, 190-204. http://doi.org/10.1016/j.isprsjprs.2018.06.008.

[2] **Hong, F.**, Zhan, W. et al., Fast generation of physically true daily land surface temperatures using thermal observations from tandem polar-orbiters, Part I: Framework and validation, in preparation. (Target Journal: *Remote Sensing of Environment*)

[3] **Hong, F.**, Zhan, W. et al., Fast generation of physically true daily land surface temperatures using thermal observations from tandem polar-orbiters, Part II: Application to global scale and sampling bias analysis, in preparation. (Target Journal: *Remote Sensing of Environment*)

[4] Liu, Z., Zhan, W., Lai, J., **Hong, F.**, Quan, J., Bechtel, B., Huang, F., Zou, Z. (2019). Balancing prediction accuracy and generalization ability: A hybrid framework for modelling the annual dynamics of satellite-derived land surface temperatures. *ISPRS Journal of Photogrammetry and Remote Sensing*, 151, 189-206. http://doi.org/ 10.1016/j.isprsjprs.2019.03.013.

[5] Zou, Z., Zhan, W., Liu, Z., Bechtel, B., Gao, L., **Hong, F.**, Huang, F., Lai, J. (2018). Enhanced modeling of annual temperature cycles with temporally discrete remotely sensed thermal observations. *Remote Sensing*, 10(4), 650. https://doi.org/10.3390/rs10040650.

[6] Lai, J., Zhan, W., Huang, F., Voogt, J., Bechtel, B., Allen, M., Peng, S., **Hong, F.**, Liu, Y., Du, P. (2018). Identification of typical diurnal patterns for clear-sky climatology of surface urban heat islands. *Remote Sensing of Environment*, 217, 203-220. http://doi.org/10.1016/j.rse.2018.08.021.

Conferences

- The IEEE International Geoscience and Remote Sensing Symposium 2019, Yokohama, Japan (Attendance) Aug 2019
- 4th Quantitative Remote Sensing Forum, Nanjing, China (Poster presentation) Jun 2019
- The American Geophysical Union 2018 Fall Meeting, Washington D.C., United States (Poster presentation) Dec 2018
- 2rd Thermal Infrared Quantitative Remote Sensing Symposium, Urumqi, China (Attendance) Aug 2018
- The International Society for Photogrammetry and Remote Sensing Geospatial Week 2017, Wuhan, China (Attendance) Sep 2017
- 3rd Quantitative Remote Sensing Forum, Beijing, China (Attendance) Aug 2017

Awards and Honors

- Best poster award of 4th quantitative remote sensing forum (Ranking: 4/150) 2019
- Huawei Scholarship (2 out of 113) 2018
- Outstanding postgraduate students in Nanjing University (TOP 10%) 2018
- National Encouragement Scholarship in Nanjing University (TOP 20%) 2014 - 2016

Skills and Others

- Computer: Python, MATLAB, ArcGIS, ENVI, OriginLab, and MS Office
- Language: Fluent in English, TOEFL: 103 (R28 L27 S19 W29), GRE 152 (V) + 170 (Q) + (AW 4.0)
- Hobby: 7-year experience in playing tennis